

SQL

CTEs and Its Benefits



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1

What is a CTE?

CTE is the short form for **Common Table Expressions**.

It is one of the most powerful tools of SQL.



2

CTEs can be referenced multiple times in a single query.

This is particularly useful when you need to use the same subquery multiple times within a query.



3

To create a CTE:

- Use the **WITH** keyword to start.
- Give your CTE a name.
- Inside parentheses, define your CTE by writing a regular **SELECT** statement.
- In the main query, reference the CTE by its name.



4

Here's a simple syntax example:

```
WITH CTE_Name AS (  
    SELECT column1, column2  
    FROM your_table  
    WHERE condition  
)  
SELECT *  
FROM CTE_Name;
```



5

Benefits of using CTEs

- **Readability:** It makes complex queries easier to read and understand.
- **Structure:** It improves the structure of your SQL code by defining subqueries at the start.
- **Performance:** It can make your SQL queries run faster.



- **Manageability:** It makes your code easier to manage by defining reusable subqueries.
- **Recursive Queries:** It is great for working with hierarchical data structures. It can be used to create recursive queries, which are queries that refer to themselves.

CTEs Vs Subqueries



8

Similarities

Both are used to embed queries within another query to manipulate data.



Definition

CTE: Defined at the beginning with **WITH** keyword and given a name.

Subquery: Nested directly within the main query (no separate definition).



10

Reuse

CTE: Can be referenced multiple times within the main query.

Subquery: Can only be used once.



In Summary

CTE is a handy tool in SQL for **simplifying** queries, making them **faster**, and improving **readability**, which is crucial for working with large datasets effectively.

**I hope you find this
valuable! 🥰**

**A like from you would be a great
encouragement**



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